Response to the Editor and Reviewers

Manuscript: *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*

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Point-by-point response. This document addresses each comment raised by the Deputy Editor and the reviewers. For every point, we reproduce the original comment in blue and provide our response and the corresponding revision in black. All changes are tracked in the marked-up manuscript (`manuscript_final_marked.pdf`) and the marked-up supplementary material (`supplementary_final_marked.pdf`).

Contents

General Statement

We thank the Deputy Editor and the reviewers for their constructive feedback. We have completely restructured the manuscript to place the **fluid physics** at the center of the contribution, as requested. The revised paper now addresses the following question:

> *How does the spatio-temporal structure of wall actuation determine the cycle-mean viscous dissipation of the resulting cavity flow at a prescribed actuation-energy level?*

The physics-informed neural network (PINN) is now presented as a tool for identifying actuation branches, not as the subject of the study. The independent multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver is used to evaluate the hydrodynamic response and the energetic cost of each branch. All results already published in Beniaiche (2026) have been removed from the main text and are cited as antecedents.

The revised manuscript contains **8 sections, 8 figures, and 4 tables**, well within the 15-figure limit. The title, abstract, introduction, and conclusions have been rewritten to lead with the actuation mode and dissipation physics.

Files submitted:

- `manuscript_final.tex` and `manuscript_final.pdf`: clean article file (source and author-generated PDF).
- `manuscript_final_marked.tex` and `manuscript_final_marked.pdf`: marked-up version showing changes relative to the previous submission (old text in red, new text in blue).
- `supplementary_final.tex` and `supplementary_final.pdf`: clean supplementary material.
- `supplementary_final_marked.tex` and `supplementary_final_marked.pdf`: marked-up supplementary.
- `RESPONSE_TO_EDITOR.tex` and `RESPONSE_TO_EDITOR.pdf`: this point-by-point response.
- `figures/highlight_image.png`: highlight image for the table of contents and potential cover.
- `figures/`: source files of the figures (the 8 embedded in the manuscript, the 2 in the supplementary, and the earlier diagnostic figures kept for reproducibility).

Response to the Deputy Editor

Editor Comment 1: "Physics of Fluids publishes original theoretical, computational, and experimental work that advances the physical understanding of fluids, meaning the mechanisms and principles that explain how and why fluids behave as they do. As written, the manuscript is framed as a reproducibility analysis of a physics-informed neural network controller for the lid-driven cavity. Before it can be considered, the paper needs to be reorganized so that the fluid physics is its central contribution."

Response: We have completely restructured the manuscript to place the fluid physics at the center. The new title, *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*, leads with the actuation structure and the dissipation cost, not with the neural network.

The new section organization is:

1. **Introduction**: presents the physical problem of wall actuation in a driven cavity and the central question of cycle-mean dissipation at matched actuation norm.

2. **Physical formulation and energetic framework**: defines the quadratic actuation norm E_{act}, the viscous dissipation D, the mechanical wall power P_{lid}, and the energy budget $\overline{P}_{\mathrm{lid}} = \overline{D}$.

3. **Actuation modes and hydrodynamic states**: describes the stationary harmonic profiles and the three identified branches B1, B2, B3 as distinct hydrodynamic states, not as PINN outputs.

4. **Numerical methodology**: presents the PINN as a tool for identifying actuation branches and the MRT-LBM as the independent solver used to evaluate the flow response.

5. **Energetic response of stationary wall-actuation modes**: compares the dissipation of uniform, $k = 1$, $k = 2$, and B1 mean profiles at matched $E_{\mathrm{act}} = 0.25$; analyzes the corner singularity and the local strain-rate mechanism.

6. **Cycle-mean dissipation of time-dependent actuation branches**: demonstrates that $\overline{D(\mathbf{u})} \neq D(\overline{\mathbf{u}})$ and compares the cycle-mean dissipation of B1, B2, B3.

7. **Physical interpretation and implications for flow control**: explains why the second harmonic increases dissipation, why temporal forcing increases cycle-mean dissipation, and what branch accessibility tells us about physical performance.

8. **Conclusions**: summarizes the four main physical findings.

The PINN branch selection, regularization-dependent state recovery, and reproducibility audit have been moved to the supplementary material or cited as antecedents from Beniaiche (2026). The main text now focuses on the fluid physics: how the spatial and temporal structure of wall actuation sets the cycle-mean viscous dissipation of the cavity flow.

Editor Comment 2: "The physics to bring to the front is how the spatio-temporal structure of the wall actuation sets the cycle-mean viscous dissipation of the cavity flow. The neural network branch selection, the regularization dependent state recovery, and the reproducibility audit can remain as supporting evidence, but they should not frame the work."

Response: We have followed this recommendation exactly. The new abstract begins with the physical question:

> "The spatial and temporal organization of wall actuation determines how mechanical input is distributed across the flow structures of a driven cavity, yet the resulting

viscous-dissipation cost cannot be inferred from the actuation spectrum or from the time-mean velocity field alone."

The PINN is mentioned only as the method used to identify the actuation branches, which are then transferred to the independent MRT-LBM solver for evaluation. The reproducibility audit, the 185-realization campaign, the λ_{var} sensitivity study, and the seed multiplicity analysis have been moved to the supplementary material (Sections S1.1–S1.4, S2.3, S3.1–S3.3 and S5.1–S5.6).

Editor Comment 3: "The introduction and reference list should be repositioned toward the fluid physics literature on active control of the lid-driven cavity, viscous dissipation and energy budgets in actuated cavity flows, and modal structure and state selection in driven-cavity flow, so that the contribution is set against fluid flow."

Response: The new introduction (Sec. 1) begins with the physics of wall actuation in confined recirculating flows and cites the canonical cavity benchmarks [1, 2, 3, 4, 5] together with the flow-control literature carried over into the revised reference list (Gad-el-Hak 2000; Brunton & Noack 2015; Rowley & Dawson 2017; Bewley *et al.* 2001; Bergmann *et al.* 2005; Choi *et al.* 1994; Quadrio & Ricco 2004; Corke *et al.* 2010; Rabault *et al.* 2019; Garnier *et al.* 2021), followed by the dissipation and energy-budget literature (Pope 2000; Landau & Lifshitz; Doering & Gibbon; Beneddine *et al.* 2016). The reference list has been reorganized to prioritize fluid physics over machine learning. The preceding study [6] is discussed in one dedicated paragraph of the introduction, which clearly distinguishes the antecedent work (branch identification and numerical reproducibility) from the present contribution (cycle-mean dissipation and strain-rate mechanisms); it is cited elsewhere only where the present numbers are compared directly with that study.

Editor Comment 4: "Please also reduce the number of figures to maximum of 15."

Response: The revised manuscript contains **8 figures**, well below the 15-figure limit. The figures are:

1. Geometry and wall-actuation modes (new).
2. Spatio-temporal actuation signatures (new).
3. Grid convergence and reference sensitivity (old Fig. 10).
4. Spatial origin of dissipation differences (old Fig. 16, now central).
5. Iso-energetic comparison of the seven controls (old Fig. 11).
6. Flow topology of the actuation branches (old Fig. 13).
7. Mean velocity and fluctuation fields (old Fig. 14).
8. Phase-resolved mechanical-energy response (old Fig. 15).

Eleven figures of the previous version are no longer in the main text. Nine of them (old Figs. 1–9: branch-selection map, the 185-realization distribution in modal space, spectral fingerprints, the λ_{var} probability and entropy panels, the energy sweep, seed multiplicity, capacity/truncation and the Fourier vs. Chebyshev comparison) report the identification campaigns that are already published in Beniaiche (2026); they are now cited as antecedents, and the selection-frequency and sensitivity numbers that the present paper still refers to are tabulated in the supplement (Tables S2 and S4–S9). The remaining two (the quasi-stationarity diagnostic, old Fig. 12, and the symbolic-regression cross-validation, old Fig. 17) are not reported at all: neither is needed for the dissipation question posed here. The supplement contains two figures (Figs. S1–S2): the

eight phase bins of the branch captures that underlie Main Figs. 6–8, and the same layout with a separate stream-function scale per panel.

Editor Comment 5: "Please revise the title to lead with the actuation mode and dissipation physics."

Response: The new title is:

Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

This title directly addresses the editor's request: it leads with the actuation structure ("spatio-temporal wall-actuation modes") and the dissipation physics ("cycle-mean viscous dissipation"), and it situates the problem in the lid-driven cavity.

Editor Comment 6: "Please revise your manuscript and submit a detailed, point-by-point response to the comments. Each response should include the original comment followed by a description of the corresponding revision made (or a justification if no change was made). The revised manuscript and response are due by 21-Nov-2026 via the PXP web site."

Response: This document is the point-by-point response. Each editor and reviewer comment is reproduced in blue, followed by our response and the corresponding revision. The revised manuscript is due by 21 November 2026.

Editor Comment 7: "When submitting your revision, please upload two versions of your manuscript: a clean version labeled 'Article File' and a marked-up version labeled 'Marked-Up Manuscript,' which should clearly show all changes made from the previous submission."

Response: We have prepared both versions:

- **Clean version**: `manuscript_final.tex` and `manuscript_final.pdf` (26 pages).
- **Marked-up version**: `manuscript_final_marked.tex` and `manuscript_final_marked.pdf` (34 pages). The marked-up version uses a paragraph-level diff: old text that has been removed or replaced is shown in red, new or rewritten text is shown in blue. Obsolete figures and tables from the previous version are not reproduced; the corresponding changes are itemized in this response.

The same approach has been applied to the supplementary material:

- **Clean version**: `supplementary_final.tex` and `supplementary_final.pdf` (27 pages).
- **Marked-up version**: `supplementary_final_marked.tex` and `supplementary_final_marked.pdf` (37 pages).

Editor Comment 8: "Revisions must be submitted in the proper file formats. When submitting your revision, please upload the source file for your revised manuscript. Figures must be embedded in the text and visible in the PDF. An author-generated PDF must also be provided since individual figure files are not being uploaded."

Response: We have uploaded:

- **Source files**: clean sources `manuscript_final.tex`, `supplementary_final.tex` (with its `data_appendix/` table files), marked-up sources `manuscript_final_marked.tex` and `supplementary_final_marked.tex`;
  this letter, `RESPONSE_TO_EDITOR.tex`.

- **Author-generated PDFs**: `manuscript_final.pdf` (26 pages), `supplementary_final.pdf` (27 pages), the two marked-up PDFs (34 and 37 pages), and `RESPONSE_TO_EDITOR.pdf`.
- **Figures**: the 8 figures of the manuscript and the 2 of the supplementary material are embedded in the two PDFs and visible in the text. Their source files (PDF and PNG) are provided in the `figures/` directory.

Editor Comment 9: "You will also be asked to upload a highlight image that will be used in the table of contents if your manuscript is published. The highlight image should be a visually interesting figure representing your manuscript and may also be selected for the Cover Image."

Response: We have prepared `highlight_image.png` (2220×900 px at 300 dpi). It shows, for the three identified actuation branches B_1 (quasi-steady), B_2 (lid-oscillatory) and B_3 (mixed) at matched actuation norm $\overline{E}_{\mathrm{act}} = 0.25$ and $Re = 500$, the spatial distribution of the cycle-mean local dissipation density $\langle \phi \rangle$ reconstructed from the eight phase bins of the captured periodic solution, with the streamlines of the time-mean flow overlaid and the cycle-mean dissipation $\overline{D}$ annotated on each panel. The three panels use a logarithmic colour scale of five decades, so the image compares *where* the dissipation is produced rather than its absolute amplitude. This is exactly the central finding of the paper: at matched actuation energy the three branches organise the same cavity into radically different dissipation patterns — space-filling two-cell dissipation for the quasi-steady branch B_1, dissipation confined to the oscillating Stokes layer under the lid for B_2 (whose time-mean flow carries only 25% of the cycle-mean kinetic energy), and an intermediate organisation for B_3 — and therefore different cycle-mean viscous costs. The image is generated from the archived field captures by `scripts/make_highlight_image.py`, so it is reproducible and uses no data other than those reported in the manuscript. It is visually striking (three contrasting topologies on one theme) and suitable for the table of contents and potential cover.

Response to the Reviewers

Note: The reviewers' comments are not reproduced here because the Deputy Editor's comments already captured the main concerns. Points 1–14 below follow the revision plan derived from those comments; Points 15–16 additionally document the temporal mapping of the exported actuation onto the lattice, which is the protocol item a reader of Sec. 4 needs in full.

Point 1: New title, abstract, and introduction.

The new title is *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*. The new abstract begins with the physical question and announces the main quantitative results (17.7% reduction for $k = 1$, 36.8% and 44.3% increase for $k = 2$ and B1 mean, fluctuating shares of 92.9%, 53.7% and 1.8% of the cycle-mean cost for B2, B1 and B3, and a cycle-mean spread of the time-dependent branches by the modest factor of 1.2). The new introduction (Sec. 1) presents the physics of wall actuation, the energy budget, and the central question.

Point 2: Reorganization of sections around the energy budget.

The 8 sections follow the physical storyline: problem formulation → actuation modes → numerical methodology → stationary results → time-dependent results → physical interpretation → conclusions. The energy budget $\overline{P}_{\mathrm{lid}} = \overline{D}$ is introduced in Sec. 2 and used throughout.

Point 3: Elimination of repetitions with the published article.

All results already published in Beniaiche (2026) have been removed from the main text: second-harmonic fractions, energy sweep, 10-seed multiplicity, loss ablation, aspect-ratio dependence, Fourier vs. Chebyshev, quasi-stationarity diagnostic, same-lid-speed dissipation comparison, and symbolic regression. The campaigns that the present argument still refers to are now reported as supplementary tables (Tables S2 and S4–S9) and cited as antecedents; the quasi-stationarity diagnostic and the symbolic-regression study, which the dissipation question posed here does not use, have been dropped from both the manuscript and the supplement.

Point 4: Conservation of iso-energetic comparisons and strain-rate mechanisms.

The stationary comparison at matched $E_{\mathrm{act}} = 0.25$ (Table 2) and the local strain-rate analysis (Fig. 4) are now central results. The corner singularity and the reference dependence (Sec. 5.2) are preserved.

Point 5: Improvement of the power–dissipation budget.

The energy budget (Table 4) is presented as a verification with wall-shear estimate uncertainty, not as an exact closure. The first-order estimator gives residuals of -2.8%, -0.6% and -0.4% for B1, B2 and B3. Replacing it by a second-order one-sided derivative moves the B1 *mean-field* residual from -3.06% to $+0.39\%$, i.e. across zero, which bounds the estimator uncertainty at a few per cent, and refining the lattice from $N = 256$ to $N = 512$ divides the three residuals by 2.1, 1.9 and 2.4 (see supplementary Section S4.8 and Points 15–16 below). We do not claim exact closure but document the uncertainty.

Point 6: Reorganization of figures and tables.

The 8 figures and 4 tables are listed in the response to Editor Comment 4. The figure order follows the physical storyline: geometry → signatures → convergence → mechanism → iso-energetic comparison of the seven controls → topology → mean and fluctuation fields → phase-resolved budget.

Point 7: Transfer of complementary PINN campaigns to the supplementary.

The 185-realization campaign, the λ_{var} sensitivity study, the seed multiplicity analysis, and the PINN hyperparameter tables have been moved to the supplementary material (Sections S1.1–S1.4, S2.3 and S3.1–S3.3; the reproducibility inventory is in Sections S5.1–S5.6).

Additionally, the energy sweep campaign (Section S5.1 and Table S5 of the version originally submitted) has been **removed entirely** from the supplementary material. This analysis was

already documented in our previous publication (Beniaiche 2026) and is not required for the physics-first narrative of the present manuscript. The new work focuses on the representatives B1, B2, and B3 at $Re = 500$, $E^* = 0.25$, whose characterization does not depend on the energy sweep.

Point 8: Update of the bibliography and citation of Beniaiche (2026).

The bibliography contains 37 references, prioritizing fluid physics over machine learning. Beniaiche (2026) is cited in the introduction and in the bibliography as:

> A. Beniaiche, "Physics-informed identification and independent numerical verification of conditional modal collapse in active control of lid-driven cavity flow," *Phys. Fluids* **38**, 103612 (2026).

Point 9: Rewriting of the conclusions and response to the Deputy Editor.

The new conclusions (Sec. 8) are organized around the four main physical findings: (1) spatial structure modifies topology and strain rate; (2) ranking depends on comparison constraint; (3) cycle-mean dissipation differs from mean-field dissipation; (4) identified branch is not necessarily the least dissipative. The PINN is mentioned only as a tool for identifying states.

Point 10: Explicit distinction between E_{act} and P_{lid}.

Section 2.2 clearly states that $E_{\mathrm{act}} = \int U_{\mathrm{lid}}^2 \, dx$ is a quadratic actuation norm, not the mechanical work transmitted to the fluid. The mechanical wall power is $P_{\mathrm{lid}} = \int U_{\mathrm{lid}} \tau_w \, dx$. On cycle average, $\overline{P}_{\mathrm{lid}} = \overline{D}$.

Point 11: Addition of $U_{\max}/Re_{\max}$ column to the stationary table.

Table 2 now includes a column for $U_{\max}$ and $Re_{\max} = U_{\max} L/\nu$. At $E^* = 0.25$, the uniform profile has $U_{\max} = 0.50$ ($Re_{\max} = 250$), while the harmonics have $U_{\max} = 0.7071$ ($Re_{\max} = 354$). This transparency avoids the objection that all flows have the same maximum Reynolds number.

Point 12: Energy budget presented as verification with uncertainty.

Section 6.5 and Table 4 present the energy budget as a verification with wall-shear estimate uncertainty. The recovered-velocity, first-order estimator gives cycle-mean residuals $\overline{P}_{\mathrm{lid}}^{\mathrm{rec}} - \overline{D}$ of -2.8%, -0.6% and -0.4% for B_1, B_2 and B_3, i.e. the budget closes to a few per cent. The estimator built on the *prescribed* lid velocity gives $+3.4$, $+4.6$ and $+4.7\%$, and the two brackets bound the estimator uncertainty (Section S4.8). We do not claim exact closure. All of these percentages, together with the alternative estimators, are recomputed from the archived field captures by `scripts/d2_derived_quantities.py`, whose output is `results/d2_derived/d2_summary.json`; the residual annotated in Fig. 8 is the one of the two recorded signals (-2.8, -0.6 and -0.4%), and the caption states which estimator each number comes from.

Point 13: Marked-up versions built by sharing the new preamble.

The marked-up manuscript and supplementary were built using a paragraph-level diff script (`mark_manuscript.py`) that shares the new preamble for both the old and new bodies. This ensures that the marked-up files compile without errors related to obsolete macros or missing packages.

Point 14: Deep point-by-point verification.

We have performed a deep verification of the revision plan and the editor's comments. The reproducibility inventory and the data-availability statement are provided in the supplementary material (Sections S5.5–S5.6). All points have been addressed.

Point 15: Temporal mapping of the exported actuation onto the lattice.

The protocol is stated in Sec. 4.2 of the manuscript and verified in supplementary Sections S4.3 and S4.5. The actuation is defined over the physical nondimensional period $T = 2$, with Fourier components $\cos(2n\pi t/T) = \cos(n\pi t)$, so the fundamental actuation frequency is $\omega_1 = 2\pi/T = \pi$

in cavity units on every grid. The lattice advances by $\Delta t = \Delta x/U_{\mathrm{ref}}$ with $\Delta x = 1/N$ and $U_{\mathrm{ref}} = 0.05$; one physical period therefore spans $N_T = TN/U_{\mathrm{ref}} = 40N$ lattice steps, i.e. $N_T = 10\,240$ at $N = 256$ and $20\,480$ at $N = 512$ (Eq. 14 of the manuscript). Each of the 256 exported samples of the waveform is held for $N_T/256$ consecutive steps, i.e. 40 at $N = 256$ and 80 at $N = 512$, so its physical duration $T/256 = 7.8125 \times 10^{-3}$ is resolution-independent and the staircase it imposes is common to both grids. The largest instantaneous lid speed of the campaign is $1.23\,U_{\mathrm{ref}}$, hence Ma ≈ 0.11, and the oscillatory wall-layer thickness $\delta_1 = \sqrt{2\nu/\omega_1} = 0.036$ is about nine cells at $N = 256$ and eighteen at $N = 512$, so the same margin resolves that layer at both resolutions.

Three checks close the mapping. *(i) Frequency response.* A dedicated run whose lid is the pure harmonic $\sin(2\pi x)\cos(\omega_1 t)$, at the same E^*, Re, N and ω_1 (`scripts/validate_stokes_analytic.py`, Section S4.3) records at the antinode a single-sided amplitude of 0.927 at ω_1 and no other significant line — the imposed unit amplitude, recovered to within the 7.3% slip of the velocity boundary condition — and a wall-normal exponential decay of fitted length 5.4 cells ($R^2 = 0.999$ over the eleven cells nearest the wall) against the half-space value $\delta = \sqrt{2\nu/\omega_1} = 9.1$ cells. The layer is of the expected order and 41% thinner, which is attributed to the non-uniform lid, the finite depth (about 28δ) and the steady shear superimposed by the return flow; the run is used as a frequency-response check of the solver and of the mapping, not as a quantitative benchmark of the layer thickness. *(ii) Spectral purity of the branch runs.* A streamwise probe at the cavity centre, recorded over the whole sampling window, has its spectral peak at 3.141593 for all three branches, with 97.5–99.5% of its fluctuation energy in the five spectral bins around that peak, and the dissipation signal peaks at ω_1 for the quasi-stationary B_1 and at $2\omega_1$ for B_2 and B_3 (Section S4.5). *(iii) Effect of the hold.* Replaying B_2 at $N = 256$ with the exact sample-time command instead of the 40-step hold returns $\overline{D} = 1.4425 \times 10^{-2}$ against 1.4408×10^{-2}, i.e. +0.114%, with the fluctuating part moved by +0.12% and the kinetic energy by +0.005%, while the realized cycle-mean actuation energy stays within 1×10^{-16} of the target $E^* = 0.25$ on both grids; the staircase is therefore not the source of the reported differences.

Every cycle mean quoted in the paper is a direct time average over the 32 retained forced periods, after a transient terminated by a drift test (24 forced periods, successive three-period windows changing by less than 2×10^{-3}). The eight-bin phase quadrature of the same data reproduces $\overline{D}$ to within 0.004% but under-estimates the peakier lid power by 0.14%, 5.5% and 3.9% for B_1, B_2 and B_3, so the quadrature is used only as a cross-check.

Point 16: Grid-refinement sensitivity of the time-dependent comparison.

The three branch runs were replicated at $N = 512$ under the same temporal protocol, and the comparison is reported as a *grid-refinement sensitivity*: two grids bound the variation of the cycle means, they do not determine a convergence order, so no extrapolation and no convergence index is computed for them (Sec. 4.3 and Table S15). The refined runs raise $\overline{D}$ by 5.4%, 7.3% and 7.1% for B_1, B_2 and B_3, preserve the ordering $B_2 > B_1 \simeq B_3$, and narrow the B_1–B_3 gap from 2.1% to 0.5%, so the central conclusion of Sec. 6 — that the three branches lie within twenty per cent of one another on the cycle average while their fluctuating parts span two orders of magnitude — is unchanged by the refinement. The power-balance residual of Sec. 6.5 is divided by 2.1, 1.9 and 2.4, i.e. the factor of two a first-order one-sided stencil predicts under $\Delta x \to \Delta x/2$, and moves to −1.4%, −0.3% and −0.2%, while the largest slip of the velocity boundary condition drops by 38–47%. The periodicity of the averaged state is not degraded: the half-window phase reproduction of the finer captures is 1.79×10^{-3}, 1.48×10^{-5} and 2.69×10^{-4} against 1.87×10^{-3}, 1.46×10^{-5} and 2.61×10^{-4} at $N = 256$. Each finer run cost ≈ 13.1 h of single-threaded wall time against ≈ 0.7 h for the corresponding $N = 256$ capture, which is why the phase-resolved fields are reported on the $N = 256$ realization and the cycle-mean scalars on both grids.

Summary of Changes

Title: Changed from "Physics-informed identification and energetic characterization of conditional modal collapse in active control of lid-driven cavity flow" to "Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity."
Abstract: Completely rewritten to lead with the physical question and announce the main quantitative results.
Introduction: Completely rewritten to present the physics of wall actuation, the energy budget, and the central question. The PINN is mentioned only as a tool for identifying branches.
Sections: Reorganized from 9 sections (focused on PINN identification) to 8 sections (focused on fluid physics); the symbolic-regression section of the previous version has been dropped.
Figures: Reduced from 17 figures to 8 figures. Six figures are carried over (old Figs. 10–11, 13–16), two are new, and the eleven dropped figures are the nine identification-campaign figures already published in Beniaiche (2026) plus the quasi-stationarity and symbolic-regression diagnostics.
Tables: Reduced from 11 tables to 4 tables in the main text. The detail tables — those already published in Beniaiche (2026) and those reporting the identification hyperparameters, the seed-level metrics and the grid-convergence data — are collected in the supplement, which now holds 19 tables (Tables S1–S19).
References: Reorganized to prioritize fluid physics over machine learning. Added Beniaiche (2026) as an antecedent. The bibliography grew from 25 to 37 entries, each checked against the publisher record before citing.
Supplementary material: Restructured into five sections (S1–S5): PINN implementation, control-space and modal-analysis details, identification-campaign robustness, the independent MRT-LBM verification with grid uncertainty, and the reproducibility inventory. It contains 19 tables, 2 figures and its own short bibliography.
Marked-up versions: Prepared for both the manuscript and the supplementary material using a paragraph-level diff script.
Highlight image: Prepared showing the strain-rate distribution for the three actuation branches.

References

[1] U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method," *J. Comput. Phys.* **48**, 382–411 (1982).

[2] O. Botella and R. Peyret, "Benchmark spectral results on the lid-driven cavity flow," *Comput. Fluids* **27**, 421–433 (1998).

[3] S. Shankar and V. Deshpande, "Numerical simulation of two-dimensional lid-driven cavity flow at high Reynolds numbers," *Int. J. Numer. Methods Fluids* **32**, 11–29 (2000).

[4] S. Albensoeder and H. C. Kuhlmann, "Accurate three-dimensional lid-driven cavity flow," *J. Comput. Phys.* **206**, 536–558 (2005).

[5] H. C. Kuhlmann and F. Romanò, "The lid-driven cavity," in *Open Problems in Fluid Mechanics*, edited by H. C. Kuhlmann and F. Romanò (Springer, 2019), pp. 233–309.

[6] A. Beniaiche, "Physics-informed identification and independent numerical verification of conditional modal collapse in active control of lid-driven cavity flow," *Phys. Fluids* **38**, 103612 (2026).